



Bases & Dimension

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Introduction



Price Problem



\$ 70'000



?



\$ 160'000



Introduction



\$ 70'000



?



\$ 160'000

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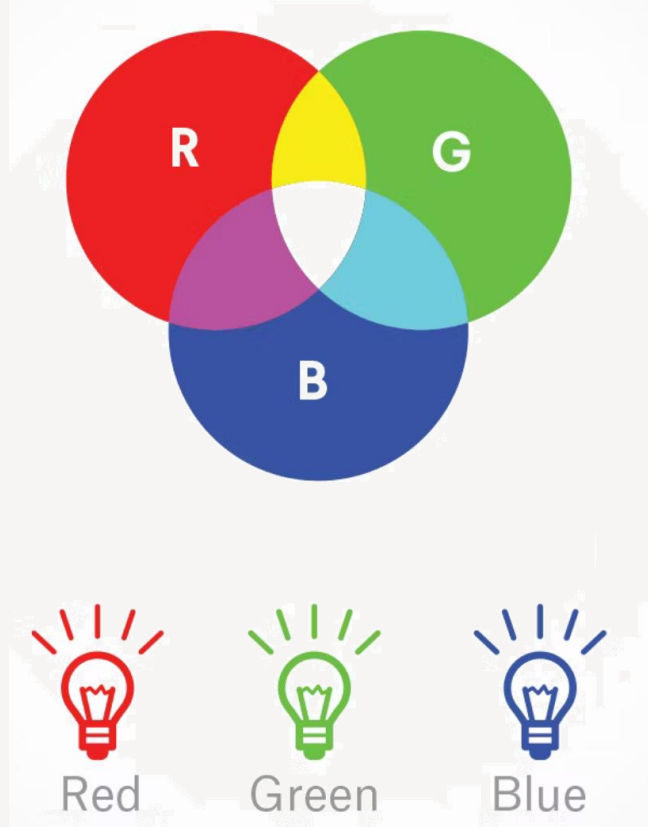
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Introduction



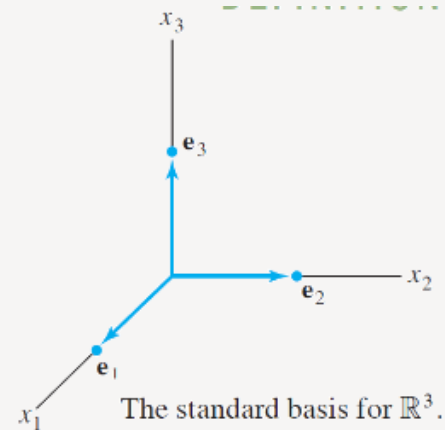
02

Basis



Basis

- A set of n linearly independent n -vectors is called a basis.
- A basis is the combination of span and independence: A set of vectors $\{v_1, \dots, v_n\}$ forms a basis for some subspace of $\mathbb{R}^n$ if it
 - (1) spans that subspace
 - (2) is an independent set of vectors.



Basis

Definition

Let H be a subspace of a vector space V . An indexed set of vectors $\mathcal{B} = \{b_1, \dots, b_n\}$ in V is a **basis** for H if

1. $\mathcal{B}$ is linearly independent set, and
2. The subspace spanned by $\mathcal{B}$ coincides with H ; that is,

$$H = \text{Span} \{b_1, \dots, b_n\}$$

Example

Which are unique?

- ☐ Express a vector in terms of any particular basis
- ☐ Bases for $\mathbb{R}^2$
- ☐ Bases with unit length for $\mathbb{R}^2$

Example Solution!

- Express a vector in terms of any particular basis

Unique: Once a basis is fixed, the coefficients (coordinates) are uniquely determined.

- 
- Bases for $\mathbb{R}^2$

Not unique: There are infinitely many possible bases, as any two linearly independent vectors can form one.

- Bases with unit length for $\mathbb{R}^2$

Not unique: There are infinitely many orthonormal bases (rotations of each other).



Vector Space of Polynomials

Be careful: A vector space can have many bases that look very different from each other!



Example (Basis)

- ☐ Standard bases for $P_n(\mathbb{R})$?
- ☐ Are $(1 - x), (1 + x), x^2$ basis for $P_2(\mathbb{R})$?

Example Solution!

- **Standard basis for $P_n(\mathbb{R})$:**

The standard basis is $\{1, x, x^2, \dots, x^n\}$.

Each polynomial in $P_n(\mathbb{R})$ can be uniquely written as a linear combination of these monomials.

- **Are $(1-x)$, $(1+x)$, x^2 a basis for $P_2(\mathbb{R})$?**

Yes. They are **linearly independent**.

$P_2(\mathbb{R})$ is **spanned** by these three vectors.

Therefore they form a valid basis for $P_2(\mathbb{R})$.

03

Dimension

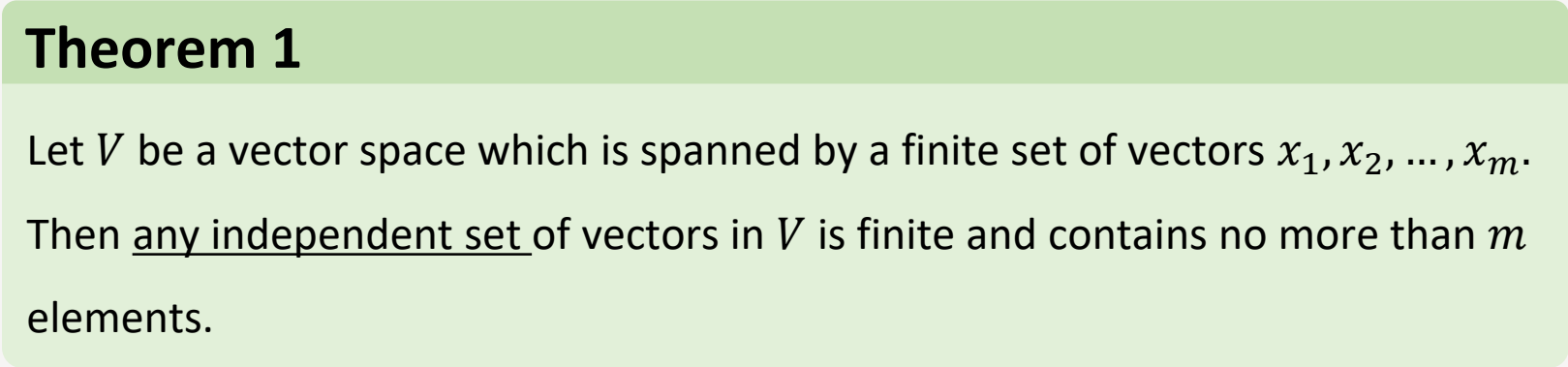


Dimensions

- The dimensionality of a vector is the number of coordinate axes in which that vector exists.
- If a vector space is spanned by a finite number of vectors, it is said to be **finite-dimensional**. Otherwise it is **infinite-dimensional**.
- The number of vectors in a basis for a finite-dimensional vector space V is called the dimension of V and denoted $\dim(V)$.

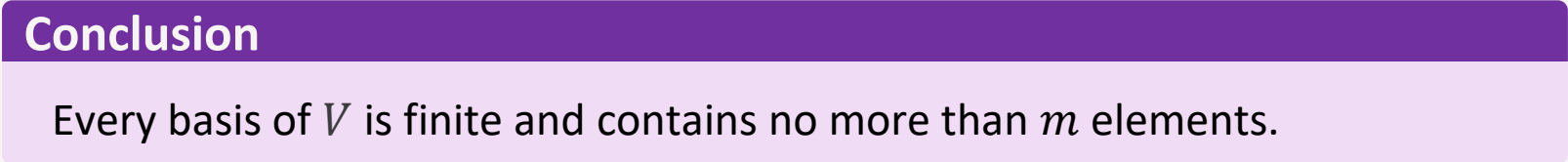
Bases and finite dimension

Theorem 1



Let V be a vector space which is spanned by a finite set of vectors $x_1, x_2, \dots, x_m$.
Then any independent set of vectors in V is finite and contains no more than m elements.

Conclusion



Every basis of V is finite and contains no more than m elements.

Proof of Theorem 1

Because $\{x_1, \dots, x_m\}$ spans V , every vector in V can be written as a linear combination of these m vectors.

Now suppose we have a linearly independent set $\{y_1, \dots, y_n\}$ in V with $n > m$. Each y_i can be expressed as

$$y_i = a_{i1}x_1 + \dots + a_{im}x_m$$

Consider a linear combination of the y_i 's that equals the zero vector:

$$0 = c_1y_1 + \dots + c_ny_n$$

Substitute each y_i in terms of the x_j 's:

$$(c_1a_{11} + \dots + c_na_{n1})x_1 + \dots + (c_1a_{1m} + \dots + c_na_{nm})x_m = 0$$

Since the x_j 's span V , this gives a homogeneous system of m equations in n unknowns $c_1, \dots, c_n$. If $n > m$, this system has a nontrivial solution (some $c_i \neq 0$), which means the y_i 's are linearly dependent. $\Rightarrow$ Contradiction!

Independent $\leq$ spanning

Conclusion

In a finite-dimensional space,

*the length of every linearly
independent list of vectors* $\leq$ *the length of every
spanning list of vectors*

Bases and finite dimension

Theorem 2

If V is a finite-dimensional vector space, then any two bases of V has the same (finite) number of elements.

Proof of Theorem 2

Let $B_1 = \{x_1, \dots, x_m\}$ and $B_2 = \{y_1, \dots, y_n\}$ be two bases of V .

Since B_1 spans V , any independent set (like B_2) can have **no more than** m elements $\Rightarrow n \leq m$.

Likewise, since B_2 spans V , any independent set (like B_1) can have **no more than** n elements $\Rightarrow m \leq n$.

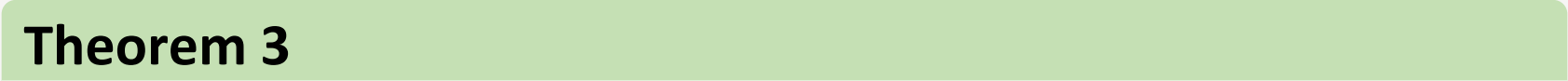
Hence $m = n$.

Basis and finite dimension

The number of vectors in a basis for a finite-dimensional vector space V is called the dimension of V and denoted as $\dim(V)$.



Theorem 2



Theorem 3

Let V be a vector space with a basis B of size m . Then

- a) Any set of more than m vectors in V must be linearly dependent, and
- b) Any set of fewer than m vectors cannot span V .

Proof of Theorem 3

- a) Since B is a basis, it spans V . By theorem 2, any linearly independent set in V can have at most m elements. Therefore, any set with more than m vectors must be linearly dependent.
- b) Suppose, for contradiction, that there exists a spanning set $\{y_1, \dots, y_k\}$ with $k < m$. From S , we can extract a basis (by removing dependent vectors). That basis would have at most k elements. But this contradicts the fact that every basis of V must have m elements.

Dimensions

Definition

A vector space V is called...

- a) **finite-dimensional** if it has a finite basis, and its **dimension**, denoted by $\dim(V)$, is the number of vectors in one of its bases.
- b) **infinite-dimensional** if it has no finite basis, and we say that $\dim(V) = \infty$.

Note

Dimension of subspace $\{\mathbf{0}\}$?

Dimensions

Example

Let's compute the dimension of some vector spaces that we've been working with.



Vector space	Basis	Dimension
$\mathcal{F}^n$ (n-tuples each elements from field $\mathcal{F}$)		
P^p (polynomials with max degree p)		
$M_{m,n}$ (matrices with m rows and n columns)		
P (all polynomials)		
F (all functions)		
C (all continues functions)		

Note!





Example. Let's compute the dimension of some vector spaces that we've been working with.

	V	Basis	$\dim(V)$
$\mathbb{R}^n, \mathbb{C}^n,$ etc. $\rightarrow$	$\mathbb{F}^n$	$\{\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n\}$	n
	$\mathcal{P}^p$	$\{1, x, x^2, \dots, x^p\}$	$p+1 \leftarrow \text{not } p!$
	$M_{m,n}$	$\{E_{1,1}, E_{1,2}, \dots, E_{m,n}\}$	mn
all polynomials $\rightarrow$	$\mathcal{P}$	$\{1, x, x^2, x^3, \dots\}$	∞
	$\mathcal{F}$	??	∞
continuous functions $\rightarrow$	$\mathcal{C}$	??	∞

04

Finite Dimensional Subspace



Basis of Subspace

Theorem 4

If W is a subspace of a finite-dimensional vector space V , every linearly independent subset of W is finite and is part of a (finite) basis for W .

Lemma

Let S be a linearly independent subset of a vector space V . Suppose u is a vector in V which is not in the subspace spanned by S . Then the set obtained by adjoining u to S is linearly independent.

Proof of Lemma

Assume

$$a_1v_1 + \cdots + a_nv_n + cu = 0$$

If $c \neq 0$, we can solve for u : $u = -\frac{1}{c}(a_1v_1 + \cdots + a_nv_n)$

This means u is a linear combination of vectors in S , i.e., $u \in \text{span}(S)$, which contradicts our assumption that $u \notin \text{span}(S)$.

Therefore, $c = 0$. The equation becomes $a_1v_1 + a_2v_2 + \cdots + a_nv_n = 0$. Since S is linearly independent, this implies $a_1 = a_2 = \cdots = a_n = 0$.

Thus, all coefficients are zero, so $S \cup \{u\}$ is linearly independent.

Proof of Theorem 4

Let $\dim V = m$. Every linearly independent set in V has at most m elements. Hence any linearly independent subset of W also has size $\leq m$, so it is finite. Therefore, the possible sizes of independent subsets of W lie in the finite set $\{0, 1, \dots, m\}$.

So largest possible size exists. Choose an independent subset $B \subseteq W$ that has this maximal size.

We claim B spans W . If not there exists a $w \notin \text{span}(B)$.

By lemma, $B \cup \{w\}$ must be linearly independent, contradicting maximality of B . Thus B is a finite basis for W .

Basis of Subspace

Corollary

A subspace is called a **proper subspace** if it's not the entire space, so $\mathbb{R}^2$ is the only subspace of $\mathbb{R}^2$ which is not a proper subspace

If W is a **proper subspace** of a finite-dimensional vector space V , then W is finite-dimensional and $\dim(W) < \dim(V)$

Corollary

In a finite-dimensional vector space V , every non-empty linearly independent set of vectors is part of basis.

Basis of sum of subspaces

Theorem 5

If W_1 and W_2 are finite-dimensional subspaces of a vector space V ,
the $W_1 + W_2$ is a finite-dimensional and

$$\dim(W_1) + \dim(W_2) = \dim(W_1 \cap W_2) + \dim(W_1 + W_2)$$

Proof of Theorem 5

Let $v_1, \dots, v_m$ be a basis of $W_1 \cap W_2$, so $\dim(W_1 \cap W_2) = m$.

Extend this basis to a basis of W_1 : $v_1, \dots, v_m, u_1, \dots, u_j$, so $\dim(W_1) = m + j$.

Similarly, extend $v_1, \dots, v_m$ to a basis of W_2 : $v_1, \dots, v_m, w_1, \dots, w_k$, so $\dim(W_2) = m + k$.

We will show that $v_1, \dots, v_m, u_1, \dots, u_j, w_1, \dots, w_k$ is a basis of $W_1 + W_2$.

1. Spanning:

Every vector in W_1 is a linear combination of v 's and u 's, and every vector in W_2 is a linear combination of v 's and w 's. Hence the span of all these vectors contains both W_1 and W_2 , so it equals $W_1 + W_2$.

Proof of Theorem 5 (cont.)

2. Linear Independence:

Suppose $a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j + c_1w_1 + \cdots + c_kw_k = 0$.

Move the w -terms to one side:

$$-(a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j) = c_1w_1 + \cdots + c_kw_k.$$

The right-hand side is in W_1 , and the left-hand side is in W_2 ; hence this vector is in $W_1 \cap W_2$, which is spanned by $v_1, \dots, v_m$. Thus

$$d_1v_1 + \cdots + d_mv_m = c_1w_1 + \cdots + c_kw_k \text{ for some } d_i\text{'s.}$$

Since $v_1, \dots, v_m, u_1, \dots, u_j, w_1, \dots, w_k$ are linearly independent all c_i 's (and d_i 's) are 0, so we have: $a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j = 0$ which implies all a_i 's and b_i 's 0 since these vectors are basis of W_1 . Hence all vectors are linearly independent.

Basis of sum of subspaces

Theorem 6

If W_1 , W_2 and W_3 are finite-dimensional subspaces of a vector space V , then can we have the following relation?

$$\begin{aligned} & \dim(W_1 + W_2 + W_3) \\ &= \dim(W_1) + \dim(W_2) + \dim(W_3) - \dim(W_1 \cap W_2) \\ & \quad - \dim(W_2 \cap W_3) - \dim(W_1 \cap W_3) + \dim(W_1 \cap W_2 \cap W_3) \end{aligned}$$

Counterexample: $W_1 = \text{span}\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right\}$, $W_2 = \text{span}\left\{\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$, $W_3 = \text{span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$

Basis of sum of subspaces

Theorem 6

If W_1 , W_2 and W_3 are finite-dimensional subspaces of a vector space V , then:

$$\begin{aligned} & \dim(W_1 + W_2 + W_3) \\ & \leq \dim(W_1) + \dim(W_2) + \dim(W_3) - \dim(W_1 \cap W_2) \\ & \quad - \dim(W_2 \cap W_3) - \dim(W_1 \cap W_3) + \dim(W_1 \cap W_2 \cap W_3) \end{aligned}$$



Problem 24: Let W_1, W_2, W_3 be subspaces of a F.D.V.S. Show that

$$\begin{aligned} \dim (W_1 + W_2 + W_3) &\leq \dim W_1 + \dim W_2 + \dim W_3 - \dim (W_1 \cap W_2) \\ &\quad - \dim (W_1 \cap W_3) - \dim (W_2 \cap W_3) + \dim (W_1 \cap W_2 \cap W_3). \end{aligned}$$

Solution: We have

$$\begin{aligned} \dim (W_1 + W_2 + W_3) &= \dim W_1 + \dim (W_2 + W_3) - \dim (W_1 \cap (W_2 + W_3)) \\ &= \dim W_1 + \dim W_2 + \dim W_3 - \dim (W_2 \cap W_3) \\ &\quad - \dim (W_1 \cap (W_2 + W_3)) \\ &\leq \dim W_1 + \dim W_2 + \dim W_3 - \dim (W_2 \cap W_3) \\ &\quad - \dim (W_1 \cap W_2) - \dim (W_1 \cap W_3) + \dim (W_1 \cap W_2 \cap W_3) \\ \text{as } (W_1 \cap W_2) + (W_1 \cap W_3) &\subseteq W_1 \cap (W_2 + W_3). \end{aligned}$$



Prob 3. Does the ‘inclusion-exclusion formula’ hold for three subspaces, i.e., is it always true that

$$\begin{aligned} \dim(U_1 + U_2 + U_3) &= \dim(U_1) + \dim(U_2) + \dim(U_3) \\ &\quad - \dim(U_1 \cap U_2) - \dim(U_1 \cap U_3) - \dim(U_2 \cap U_3) \\ &\quad + \dim(U_1 \cap U_2 \cap U_3)? \end{aligned}$$

Prove this formula or provide a counterexample.

Counterexample. Here is a fairly straightforward counterexample. Let $V = \mathbb{R}^2$, $u_1 := (1, 0)$, $u_2 := (0, 1)$, $u_3 := (1, 1)$, and let $U_j := \text{span}(u_j)$. Each U_j is 1-dimensional, being a span of a single non-zero vector. Next,

$$U_1 + U_2 + U_3 = \text{span}(u_1, u_2, u_3) \supseteq \text{span}(u_1, u_2) = V.$$

Note that none of the vectors u_j is in fact a scalar multiple of another, so any two of these vectors are linearly independent. Hence the zero vector can be represented only as a trivial linear combination of any pair of these vectors. By 1.44, this implies that

$$U_1 \cap U_2 = U_1 \cap U_3 = U_2 \cap U_3 = \{0\}.$$

Finally, $U_1 \cap U_2 \cap U_3$ is a subspace of, say, $U_1 \cap U_2$, which is the zero subspace. Hence

$$U_1 \cap U_2 \cap U_3 = \{0\}$$

as well. But then

$$\begin{aligned} \dim(U_1 + U_2 + U_3) &= 2 \neq 3 = \dim(U_1) + \dim(U_2) + \dim(U_3) - \dim(U_1 \cap U_2) \\ &\quad - \dim(U_1 \cap U_3) - \dim(U_2 \cap U_3) + \dim(U_1 \cap U_2 \cap U_3). \end{aligned}$$

Which vector spaces have bases?

Theorem 7

Let V be a finite dimensional vector space and let W be a subspace of V . Then W has a finite basis.

Theorem 8

Let V be a vector space which has a finite spanning set. Then V has a finite basis.



Which vector spaces have bases?

Without addressing this issue, here are two theorems which say that, in finite dimensions, there is no issue:

Theorem Let V be a finite dimensional vector space and let W be a subspace of V . Then W has a finite basis.

Theorem Let V be a vector space which has a finite spanning set. Then V has a finite basis.



Theorem Let V be a finite dimensional vector space and let W be a subspace of V . Then W has a finite basis.

Proof Note that every linearly independent subset of W has size $\leq \dim V$. So we can find a linearly independent subset B in W of greatest possible size.

We claim that B is a basis of W . We built B to be linearly independent, so the challenge is to show that it spans.

Let $\vec{w}$ be in W . If $\vec{w}$ is in B , then clearly $\vec{w}$ is in $\text{Span}(B)$.

If $\vec{w}$ is not in B , then $B \cup \{\vec{w}\}$ must be linearly dependent. Say

$$a\vec{w} + \sum c_i \vec{b}_i = \vec{0}$$

for $\vec{b}_i$ in B . Moreover, a must be nonzero, as B is linearly independent. Then

$$\vec{w} = -\frac{1}{a} \sum c_i \vec{b}_i$$

so $\vec{w}$ is in $\text{Span}(B)$, as required. $\square$

Theorem Let V be a vector space which has a finite spanning set. Then V has a finite basis.

Proof: Let S be the finite spanning set. Let B be the largest linearly independent subset of S ; since S is finite, this makes sense.

As before, we claim that B is a basis. As before, we just need to check that it spans.

First, we show that S is in $\text{Span}(B)$. Let $\vec{s}$ be in S . As before, if $\vec{s}$ is in B , then obviously $\vec{s} \in \text{Span}(B)$. And, if $\vec{s}$ is not in B , then $B \cup \{\vec{s}\}$ is linearly dependent and we use this to show that $\vec{s}$ is in $\text{Span}(B)$.

So S is in $\text{Span}(B)$, and $\text{Span}(S) = V$. But then B spans V , from our lemma from before. $\square$



Remark: These last two theorems were one of the first times we really needed to divide.

Suppose that we tried to do linear algebra with our scalars being the integers $\mathbb{Z}$. Let $V = \mathbb{Z}^2$, and consider $\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ and $\vec{v}_3 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$. Then one can check that $\vec{v}_1$, $\vec{v}_2$ and $\vec{v}_3$ span $\mathbb{Z}^2$. (Hint: $\begin{bmatrix} 1 \\ 0 \end{bmatrix} = 3\vec{v}_1 - \vec{v}_2 - \vec{v}_3$, $\begin{bmatrix} 0 \\ 1 \end{bmatrix} = -2\vec{v}_1 + \vec{v}_2 + \vec{v}_3$.) They are not linearly independent, since $6\vec{v}_1 = 3\vec{v}_2 + 2\vec{v}_3$.

However, no two of the $\vec{v}_i$ span $\mathbb{Z}^2$! If $\begin{bmatrix} x \\ y \end{bmatrix}$ is in $\text{Span}(\vec{v}_1, \vec{v}_2)$ then $x - y$ is divisible by 2; if $\begin{bmatrix} x \\ y \end{bmatrix}$ is in $\text{Span}(\vec{v}_1, \vec{v}_3)$ then $x - y$ is divisible by 3 and, if $\begin{bmatrix} x \\ y \end{bmatrix}$ is in $\text{Span}(\vec{v}_2, \vec{v}_3)$, then x is even and y is divisible by 3. So none of these pairs spans all of $\mathbb{Z}^2$.

This is a first example to show that linear algebra over a field is much nicer than linear algebra over what is called a ring. If you like studying hard, interesting, things, take commutative algebra! If you like studying easy, useful things, study linear algebra!

Dimensionality and Properties of Bases

Note

Let V be a finite dimensional vector space over field F . Below are some properties of bases:

1. Any linearly independent list can be extended to a basis (a maximal linearly independent list is spanning).
2. Any spanning list contains a basis (a minimal spanning list is linearly independent).
3. Any linearly independent list of length $\dim V$ is a basis.
4. Any spanning list of length $\dim V$ is a basis.

We will learn about change of basis later.

05

Coordinates



Ordered basis

Definition

If V is a finite-dimensional vector space, an **ordered basis** for V is a finite **sequence** of vectors which is linearly independent and spans V .

Be careful: The order in which the basis vectors appear in B affects the order of the entries in the coordinate vector. This is kind of janky (technically, sets don't care about order), but everyone just sort of accepts it.

Coordinate Systems

- The main reason for selecting a basis for a subspace H ; instead of merely a spanning set, is that **each vector in H can be written in only one way as a linear combination of the basis vectors.**

Note

Suppose the set $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ is a basis for a subspace H . For each x in H , the **coordinates of x relative to the basis $\mathcal{B}$** are the weights $c_1, \dots, c_p$ such that $x = c_1 \mathbf{b}_1 + \dots + c_p \mathbf{b}_p$, and the vector in $\mathbb{R}^p$

$$[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$$

is called the **coordinate vector of x (relative to $\mathcal{B}$)** or the $\mathcal{B}$ -coordinate vector of x .

Coordinate Systems

Example

Coordinate vector of $p(x) = 4 - x + 3x^2$ respect to basis $\{1, x, x^2\}$



Example Solution!

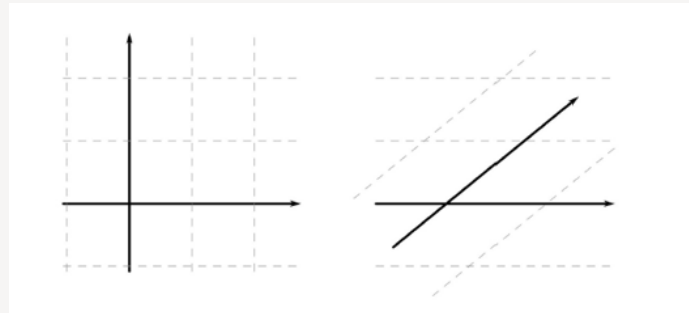
$$p(x) = 4(1) - 1(x) + 3(x^2)$$

$$[p(x)]_B = \begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix}$$



Coordinate axes

- ❑ The familiar Cartesian plane (left) has orthogonal coordinate axes. However, axes in linear algebra are not constrained to be orthogonal (right), and non-orthogonal axes can be advantageous.



Resources

- ❑ Page 97 LINEAR ALGEBRA: Theory, Intuition, Code
- ❑ Page 213: David Cherney,
- ❑ Page 54: Linear Algebra and Optimization for Machine Learning

